Examiner only

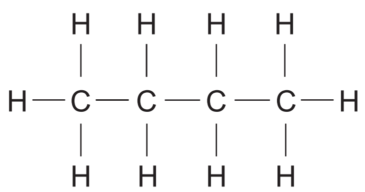
7. *(a)* The table shows the first four members of the alkane family.

Alkanes
methane, CH_4
ethane, C_2H_6
propane, C_3H_8
butane, C_4H_{10}

Give the **general** formula for the alkane family. [1]

....

(b) C_4H_{10} has two isomers. The diagram below shows the structure of one of the isomers, butane.



Draw and name the other isomer. [2]

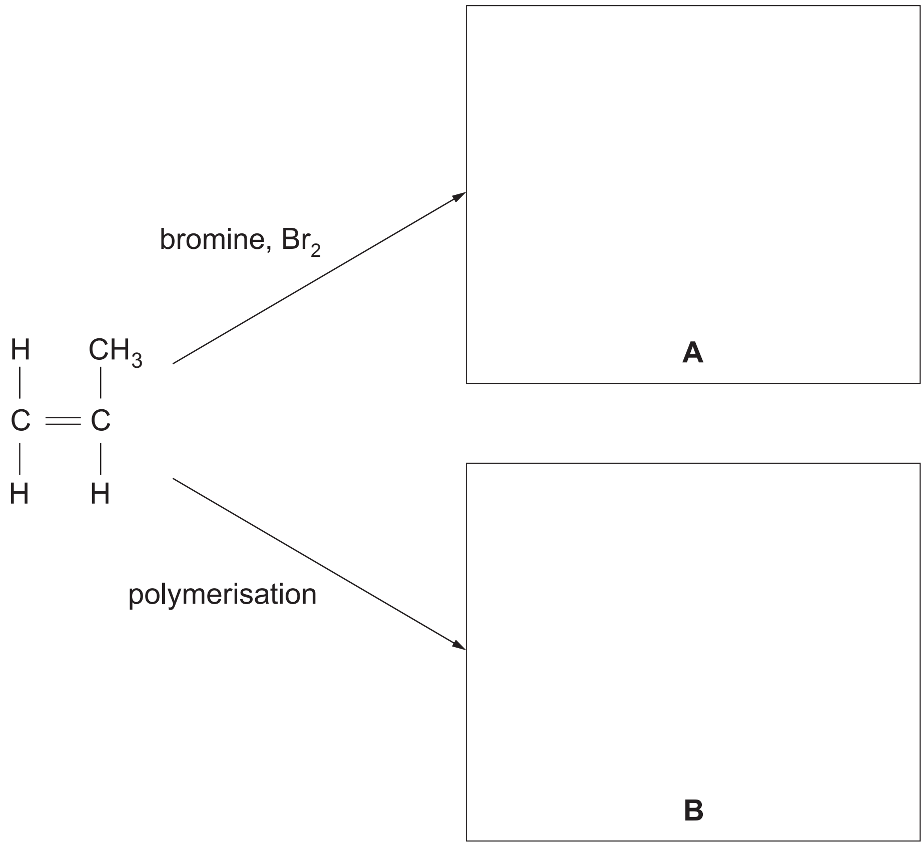
Isomer structure

Name of isomer ....

Examiner only

(c) The flow diagram shows two reactions of propene.

Draw the structural formula for compound **A** and the repeating unit in polymer **B**. [2]



(d) A student was given a colourless liquid. He suspects the liquid is ethanol.

Describe a chemical test that he could carry out to positively identify the liquid as ethanol. Include the observation he would make. [2]

Test ....

Observation ....

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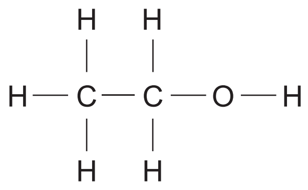
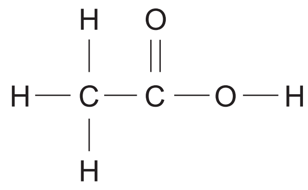
(e) Organic compounds can also be identified using infrared spectroscopy.



ethanol



ethanoic acid

Bond	Wavenumber (cm^{-1})
C═O	1650 to 1750
C—H	2800 to 3100
O—H	2500 to 3550

The infrared spectra of ethanol, ethanoic acid and one other compound are shown on the opposite page.

Use the information in the table to identify the spectra belonging to ethanol and ethanoic acid. [2]

Ethanol ....

Ethanoic acid ....

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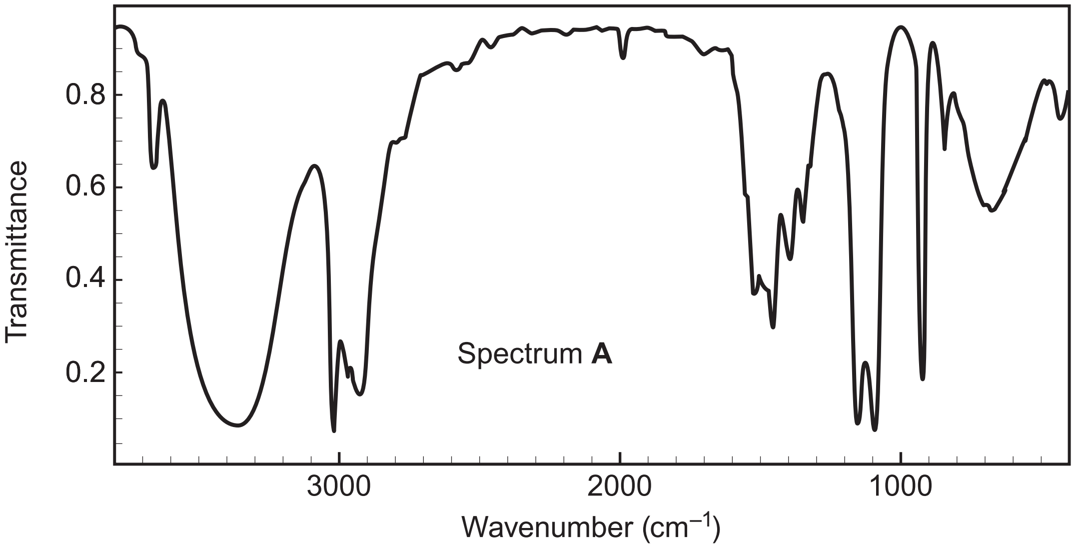
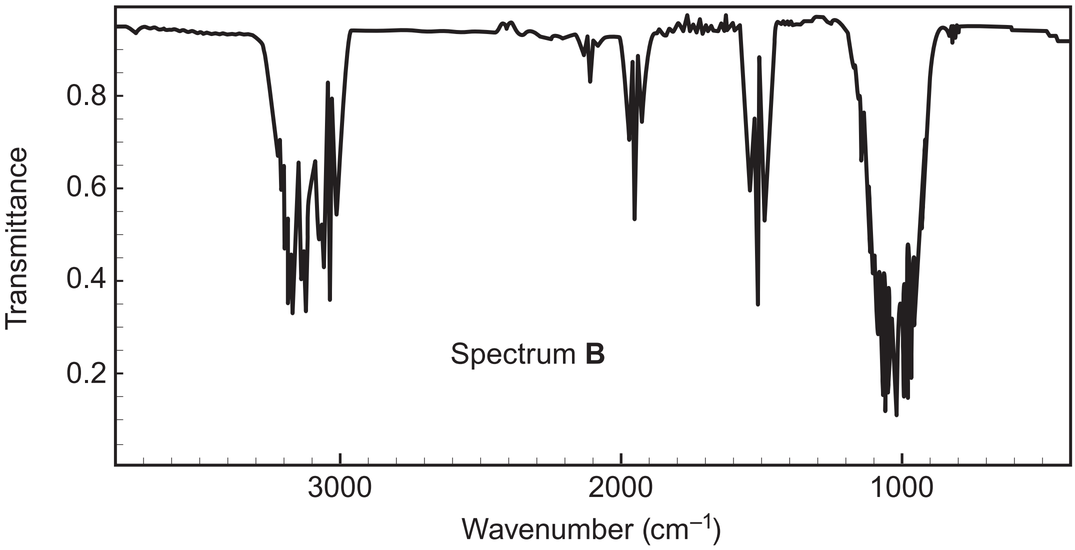
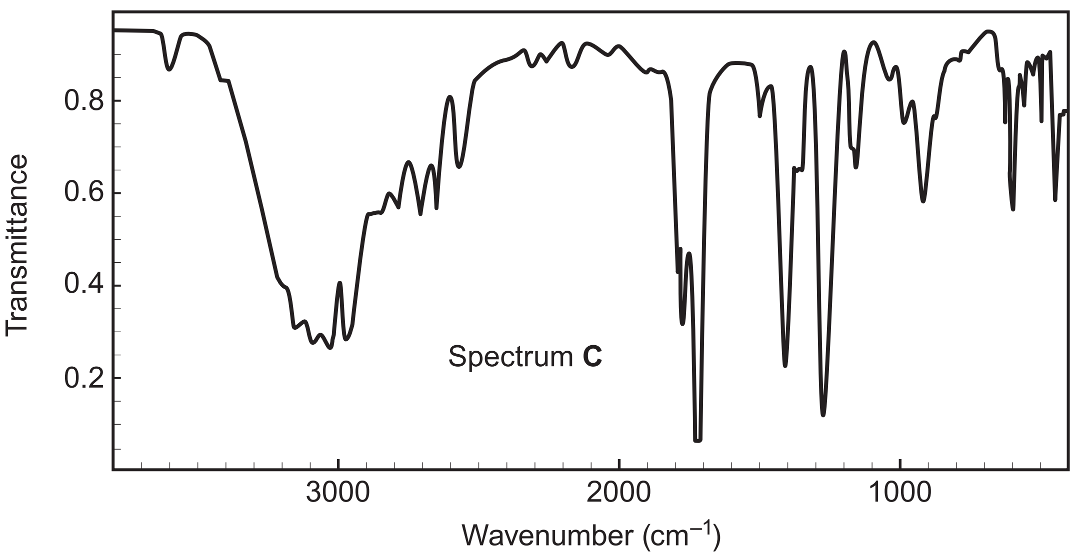


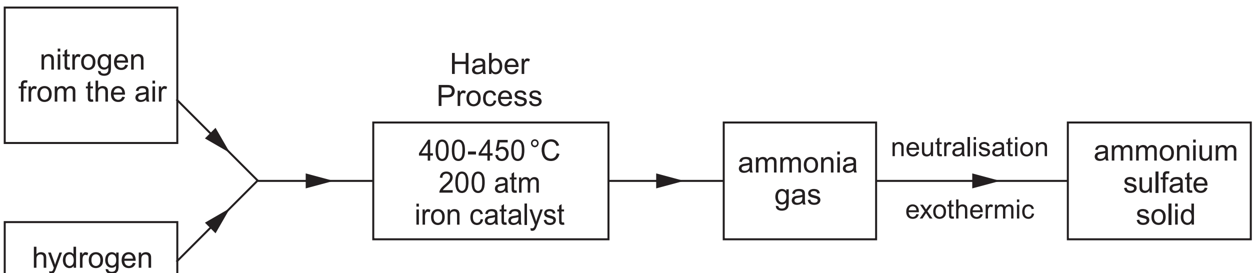




9

Examiner only

7. The flow diagram outlines the manufacture of ammonia by the Haber Process followed by the production of ammonium sulfate by neutralisation.



$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

forward reaction is exothermic

(a) Explain the choice of temperature used in the Haber Process and the reason why a catalyst is used. [3]

....

....

....

....

(b) Powdered ammonium sulfate is industrially formed by spraying sulfuric acid into a reaction chamber filled with ammonia gas. The exothermic reaction that occurs evaporates all water present in the system, forming a powdery salt.

Write a balanced symbol equation for this reaction. [2]

....

Examiner only

(c) In the laboratory ammonium sulfate is prepared by reacting ammonium hydroxide solution with dilute sulfuric acid.

Describe the titration method for making pure crystals of ammonium sulfate in the laboratory. Explain each stage of your method. [6 QER]

11

7. **Scientists make 'new fuel' discovery**

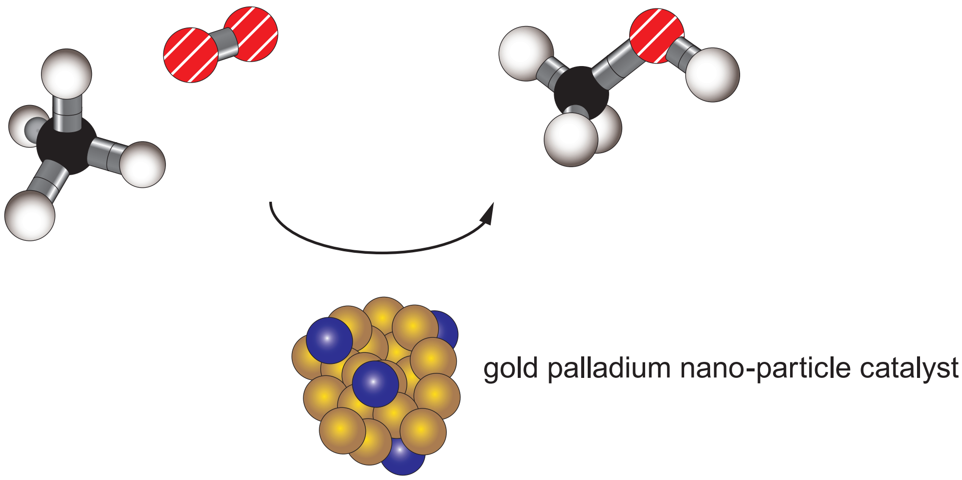
Methanol is vital in manufacturing a wide range of fuels and chemicals. Methanol is produced from methane found in natural gas. At the moment methane is condensed into liquid natural gas at the site where it is extracted and transported in pressurised containers.

Traditionally, methanol is created by converting methane into hydrogen and carbon monoxide molecules at high temperature, then rearranging the atoms in a different order in a second highly pressurised process. The current two-stage process is very energy intensive, as it burns a lot of fossil fuel to achieve high temperatures.

Stage 1 $2CH_4 + O_2 \rightleftharpoons 2CO + 4H_2$

Stage 2 $CO + 2H_2 \rightleftharpoons CH_3OH$

Scientists have discovered a new way of creating greener and cheaper methanol from methane using gold palladium nano-particles to initiate a one-stage chemical reaction that can be done at temperatures no higher than 50 °C. In this new process the gold palladium nano-particles act as a catalyst enabling methane and oxygen to combine, forming methanol in a single stage reaction.

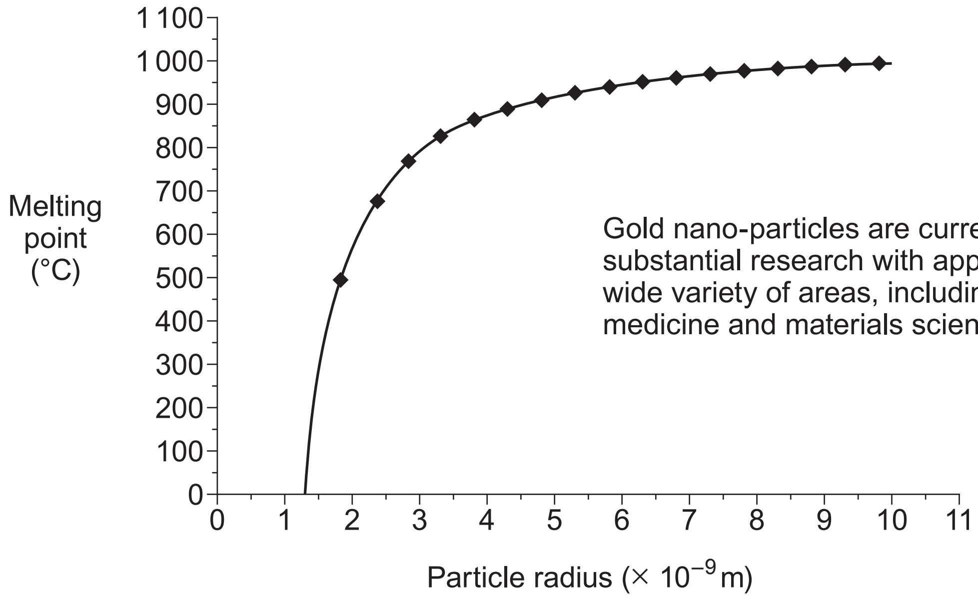


The discovery opens up the prospect of easily converting methane into methanol at the site where the methane is extracted, so that methanol can be transported as a liquid at atmospheric pressure.

Nano-particles have very different and unique properties from their bulk form.

Bulk gold	**Gold nano-particles**
• shiny	• found in a range of colours (100 nm purple; 20 nm red; 1 nm yellow)
• always gold colour	• never gold colour
• inert	• very good catalysts
• conducts electricity	• semi-conductors
• melting point 1 064 °C	• range of melting points

Examiner only



Gold nano-particles are currently subject to substantial research with applications in a wide variety of areas, including electronics, medicine and materials science

(a) Tick (✓) the box next to the **two** statements which support the opinion that the new method is more **environmentally friendly**. [2]

It is cheaper than the traditional method	☐
It uses less energy	☐
It reduces carbon dioxide emissions	☐
It uses gold nano-particles	☐
It uses more fuel	☐

(b) Complete the balanced equation for the **new** method of converting methane to methanol. [2]

$2CH_4$ + ... $\longrightarrow$ ☐ ....

Examiner only

(c) Tick (✓) the box next to the **true** statement. [1]

The melting points of gold nano-particles and bulk gold are the same ☐

Gold nano-particles have a fixed melting point value ☐

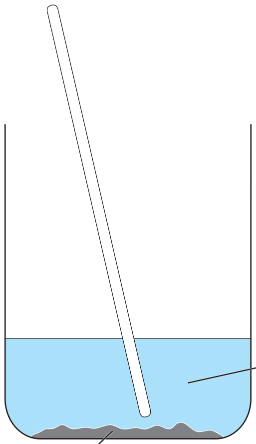
Smaller gold nano-particles have higher melting points than larger gold nano-particles ☐

The melting point of gold nano-particles depends on their size ☐

5

Examiner only

7. (a) A student investigated the mass of copper formed when magnesium powder was added to excess copper(II) sulfate solution.



The student added four different masses of magnesium powder to separate 50 cm^3 samples of copper(II) sulfate solution.

The table shows the results obtained.

Mass of magnesium added (g)	Mass of copper formed (g)
0.00	0.00
0.05	0.13
0.10	0.26
0.15	0.39
0.20	0.52

(i) Give **one** observation that would show that copper(II) sulfate is always in excess in this investigation. [1]

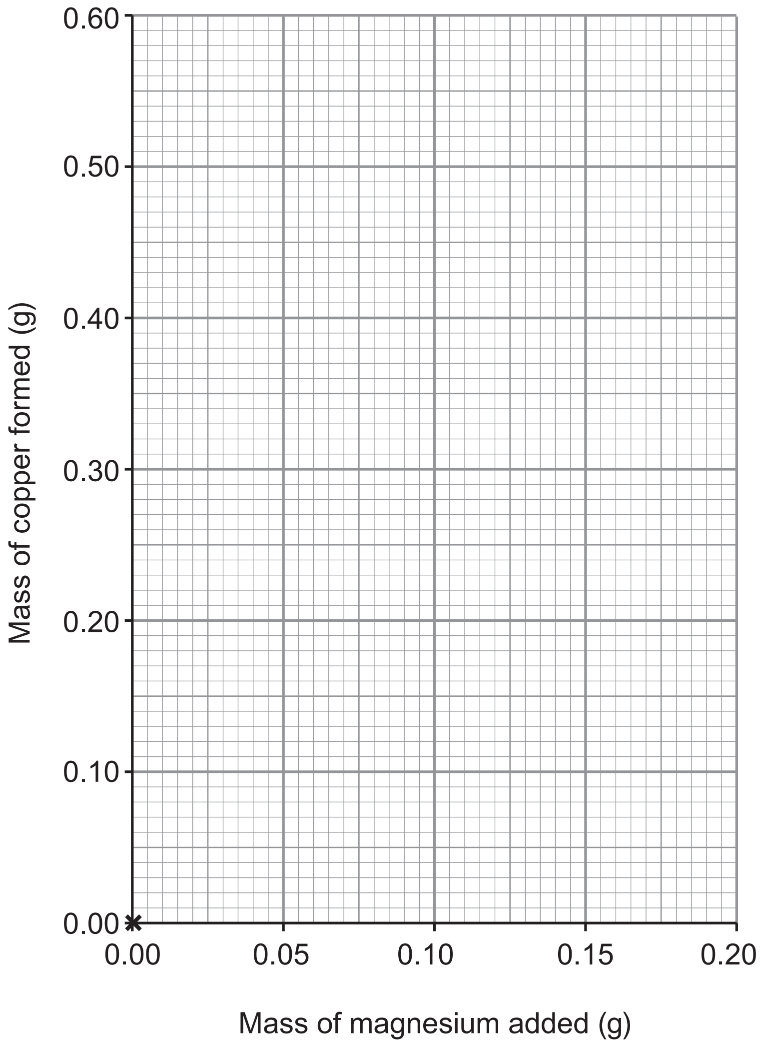
....

....

Examiner only

(ii) Plot the results from the table on the grid and draw a suitable line. [3]

The point (0,0) has been plotted for you.



(iii) I. Use the graph to find the mass of magnesium needed to form 0.18 g of copper. [1]

.... g

II. The student stated that 0.78 g of copper would be formed when 0.30 g of magnesium is used. Suggest how she arrived at this value. [1]

....

....

....

Examiner only

(b) To ensure that the copper(II) sulfate solution was always in excess the technician dissolved 3.19 g of $CuSO_4$ in 500 cm^3 of water.

(i) Calculate the number of moles of copper(II) sulfate in 3.19 g. [2]

A_r(O) = 16 A_r(S) = 32 A_r(Cu) = 63.5

Number of moles = ... mol

(ii) Use the equation below to calculate the concentration of the copper(II) sulfate solution in mol/dm^3. [2]

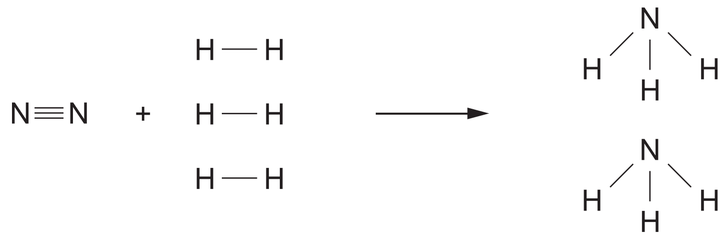
$$\text{concentration (mol/dm}^3) = \frac{\text{number of moles}}{\text{volume (dm}^3)}$$

Concentration = ... mol/dm^3

10

Examiner only

7. (a) The formation of ammonia can be represented by the following equation.



- The total amount of energy released in making the bonds in the products is 2340 kJ
- The total amount of energy released in making the bonds in the products is 94 kJ **more than** the total energy used in breaking the bonds in the reactants
- The amount of energy used to break the N≡N bond is 941 kJ

Use this information to calculate the energy used to break **one** H—H bond. [3]

Energy = ... kJ

Examiner only

(b) The Haber process used to make ammonia is usually carried out at a temperature of around 400 °C. Explain why this is the optimum temperature. [2]

A higher temperature is not used because ....

....

A lower temperature is not used because ....

....

(c) Ammonia is used in the production of fertilisers such as ammonium nitrate.

Give the balanced symbol equation for the reaction between nitric acid and ammonia to form ammonium nitrate. [2]

....

(d) Eutrophication is a problem caused by fertilisers being washed into waterways leading to overgrowth of plants. Describe how eutrophication leads to the death of aquatic organisms. [2]

....

....

....

....

9